

title: "ContentFlow: Client Portal System Manual" author: "Development Team" date: "May 2026"

1. Abstract

ContentFlow is a comprehensive content approval and management platform designed to streamline the workflow between agencies and their clients. This document serves as a technical manual and feature guide for the Client Portal. It outlines the system architecture, authentication mechanisms, real-time data synchronization, and provides a detailed breakdown of the available modules and their underlying functionalities.

2. Introduction

2.1 Purpose

The purpose of the ContentFlow Client Portal is to provide a secure, transparent, and efficient interface for clients to review, approve, and track social media content proposed by their agency. It eliminates the need for email-based approvals and consolidates communication into a single, trackable platform.

2.2 Scope

This manual covers the features accessible via the Client Portal (`/client-portal` routes), including the interactive dashboard, the real-time review interface, historical data logs, and the activity auditing system. It also touches upon the underlying technical implementations that power these features.

2.3 Technologies Used

- **Frontend Framework:** Next.js (App Router) for server-side rendering and optimized client-side routing.
- **Backend & Database:** Supabase (PostgreSQL) for relational data storage and RESTful APIs.
- **Authentication:** Supabase Auth for secure user session management.
- **Real-time Communication:** Supabase Channels for WebSocket-based real-time updates.
- **Styling:** Vanilla CSS with custom CSS variables implementing a volumetric glassmorphic design system.

3. System Architecture & Technical Overview

3.1 Authentication and Role-Based Access Control (RBAC)

The system employs a strict Role-Based Access Control mechanism. Upon logging in, the user's session is validated securely via Supabase Auth. The system queries the `users` table to determine the user's role (`admin`, `team`, or `client`) and their associated `client_id`. The Client Portal is isolated and ensures that clients can only access posts and data strictly associated with their specific `client_id`.

3.2 Real-time Data Synchronization

To ensure clients see the most up-to-date information without manually refreshing the page, the platform utilizes WebSocket connections. Components like the `AutoRefresh` module listen to specific Supabase Database channels (e.g., changes to the `posts` or `comments` tables). When a team member updates a post status or adds a comment, the payload is broadcasted, and the client's interface automatically refetches and re-renders the updated data.

3.3 Glassmorphic UI & Performance

The user interface is built using a custom "volumetric glassmorphism" design system. This relies on CSS properties like `backdrop-filter: blur()`, `rgba` background layers, and hardware-accelerated transitions to create a premium, translucent aesthetic while maintaining strict performance optimizations to avoid rendering bottlenecks on lower-end devices.

4. Module Description (Feature Guide)

4.1 Overview Dashboard

Path: `/client-portal` The Overview module serves as the primary entry point.

- **Technical Implementation:** It performs server-side data fetching to retrieve the client's profile, calculate aggregate statistics (total, pending, approved, revision counts), and fetch the queue of posts with a status of pending.
- **User Experience:** Displays a personalized greeting and a visually distinct grid of metrics. The "Pending Review" queue highlights items requiring immediate action, providing media thumbnails, platform metadata, and a direct link to the Review interface.

4.2 Post Review & Approval Workflow

Path: `/client-portal/review/[id]` This is the core module where the approval workflow occurs.

- **Interactive Platform Preview:** The system dynamically formats the provided media (`media_url`) and text (caption) to simulate exactly how it will render on the target social platform (e.g., LinkedIn, Instagram).
- **AI-Assisted Optimization Insights:** The system parses post metadata to provide calculated projections for "Expected Reach" and an "Engagement Index", helping clients make data-driven approval decisions.
- **Decision Center (State Machine):** Clients can trigger state transitions on the `posts` table:
 - **Approve:** Transitions the post status to approved.
 - **Request Changes:** Transitions the post status to revision. This action strictly requires the client to input a text payload (feedback), which is subsequently written to the `comments` table.
- **Real-time Feedback Feed:** A chronological feed querying the `comments` table. It utilizes the WebSocket connection to display new comments from admins or team members instantly.

4.3 History Archive

Path: `/client-portal/history` This module provides a comprehensive tabular view of all historical data.

- **Functionality:** It queries the entire `posts` table associated with the `client_id`.
- **Technical Features:** Implements URL-based state management (`?filter=...&sort=...`) for filtering (by status) and sorting (chronological/reverse-chronological). This allows users to bookmark specific filtered views.

4.4 Activity Log & Auditing

Path: `/client-portal/activity` A robust audit trail module designed for transparency and accountability.

- **Functionality:** Queries the `audit_log` table to present a chronological list of all system events related to the client's account.
- **Tracked Events:** Includes payloads for `post_created`, `post_approved`, `post_revision`, `post_rejected`, and `comment_added`. Each entry records the timestamp, the user ID of the actor, and contextual metadata.

4.5 Notification System

Component: `NotificationBell.js` A global module present in the application header.

- **Functionality:** Polls the `/api/notifications` endpoint to retrieve unread alerts. It acts as a centralized bell icon, visually indicating the presence of new actions requiring the client's attention, such as newly assigned posts or unread team comments.

5. Conclusion

The ContentFlow Client Portal is engineered to be both visually stunning and technically robust. By leveraging modern web technologies like Next.js and Supabase, it provides a secure, real-time, and highly efficient workflow that bridges the gap between creative teams and client stakeholders.